

Optical Diffraction-based Convolution for Semiconductor Lithography

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Abstract

*In recent years, the increasing demand for smaller and more powerful semiconductors highlighted the critical role of lithography—a key stage in semiconductor manufacturing responsible for precise mask design and wafer patterning. To meet these demands, the semiconductor industry has increasingly adopted computational lithography, employing machine learning and deep learning techniques to accelerate advancements in lithographic technology. Despite the various research efforts and successes in computational lithography, there remains a lack of explicit incorporation of physical principles. This gap limits the ability of existing methods to fully capture the complex physical phenomena inherent in lithography behaviors. To bridge this gap, we propose **OptiCo**, a novel convolutional neural network that seamlessly integrates optical diffraction principles into its architecture. At its core, OptiCo employs an optical phase kernel to model phase variations resulting from light propagation, effectively capturing the physical interactions among light, masks, and wafers. We evaluate OptiCo on semiconductor lithography benchmarks, demonstrating its superior performance in mask optimization tasks, with its remarkable generalization capabilities in OOD datasets.*

1. Introduction

Semiconductors are now indispensable to almost every facet of modern life, enabling technologies from smartphones and computers to artificial intelligence systems. Their critical role has brought semiconductor manufacturing to the forefront of technological innovation and industrial focus. Within this manufacturing ecosystem, lithography stands out as one of the most important stages [32, 55, 64]. In particular, the lithography process uses light passing through a mask to precisely project the patterns of transistors and circuits onto silicon wafers. Due to its technical demands and the precision required, the lithography stage accounts for approximately 30% of the overall manufacturing cost [29].

Over the years, the increasing demand for smaller and more powerful semiconductors has pushed lithography

technology to its physical limits. As manufacturers strive to further reduce circuit sizes, they face substantial challenges from the complex interactions between light, wafer, and mask patterns [1, 5, 15]. As shown in Figure 1, the image projected from the mask onto the wafer typically deviates from the intended design due to the effects of light diffraction, which are particularly pronounced with short-wavelength light (e.g., EUV light) used for fabricating smaller circuit feature sizes [47].

To achieve the precision in lithography, traditional approaches have relied on trial-and-error experimentation or mathematical modeling to optimize lithographic parameters [14, 30, 52, 62]. However, repeatedly executing the real lithography process is prohibitively expensive, and the complexity of light diffraction makes mathematical modeling both challenging [3, 29]. To address these constraints, the semiconductor industry has adopted simulation-based computational lithography, particularly those leveraging deep learning (DL) methods [2, 40, 46].

Among the various DL-based approaches, GAN-OPC [53] emerged as one of the earliest DL frameworks that utilize a generative adversarial network (GAN) to establish a direct mapping between input target wafer patterns and output mask designs. Building upon this foundation, DAMO [7] enhanced performance by achieving higher-resolution mask predictions by incorporating a UNet++ [66] backbone and residual blocks. TEMPO [58] further leveraged multi-domain image-to-image translation techniques to address the complexities associated with 3D mask designs.

Despite these achievements, some studies [50, 54] have highlighted limitations in their reliance on standard neural network architectures, which may struggle to capture the global information essential for accurate lithography modeling [20]. To overcome this limitation, researchers have explored the potential of Fourier Neural Operators (FNOs) [23]. Specifically, DOINN [54] introduced an FNO-based framework that employs Fourier transforms to effectively capture both low-frequency global information and high-frequency local details. Building on this approach, CFNO [50] further improved the framework by incorporating inductive lithography bias into the model architecture. While

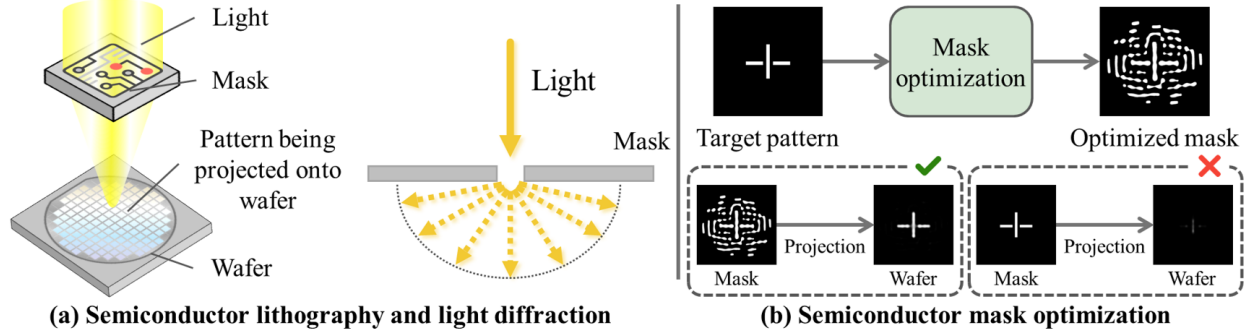


Figure 1. Illustrations of semiconductor lithography and mask optimization.

these FNO-based frameworks have demonstrated superior performance by implicitly modeling lithography in Fourier space, they did not explicitly account for the physical principles of optical diffraction theory [11, 19] that fundamentally govern lithography behavior.

To address these limitations, we propose an optical physics-inspired neural network that incorporates fundamental optical principles through the optical diffraction [4, 41]. In the context of semiconductor lithography, diffraction is particularly crucial as it determines how light propagates through the mask and ultimately forms patterns on silicon wafers [32]. A key aspect of light diffraction is the phase factor, which quantifies changes in the wavefront of light when it encounters different material interfaces [49]. For instance, as light interacts with the transparent and opaque regions of a mask, the phase factor predicts how the phase and amplitude of light waves are altered during these interactions, which influence the pattern of light projected onto wafers [13].

To model this phase factor within our neural network, we introduce an *Optical Phase (OP) kernel* that functions as a transformation layer to encode the spatial phase variations induced by light propagation. Our framework is designed to be formulation-flexible, supporting both Rayleigh–Sommerfeld-based diffraction formulations and the Hopkins TCC kernel, while remaining flexible to other kernels. Unlike traditional convolution filters that focus only on spatial features, our kernel explicitly accounts for phase variations induced by light propagation. By incorporating this physics-inspired operation into our architecture, we enhance its ability to generalize to out-of-distribution (OOD) mask designs. This capability is crucial for real-world semiconductor manufacturing, where mask patterns often deviate from the training data due to practical constraints. In addition, we adopt a simple Total Variation (TV) loss to suppress artifacts and smooth mask boundaries, further improving the manufacturability of the resulting masks.

Our main contributions can be summarized as follows:

- We propose a **Optical** diffraction-based **Convolutional** neural network (**OptiCo**) designed to effectively simulate the behavior of light for computational lithography.

- To the best of our knowledge, OptiCo is the first convolutional neural network framework to explicitly integrate the physical principles of optical diffraction into its core operations, thereby ensuring that the model inherently respects the underlying physics that govern lithography.
- We introduce a *optical phase (OP) kernel* that encodes spatial phase variations caused by light propagation. This physics-inspired and formulation-flexible OP kernel significantly improves the generalization performance for OOD mask designs, while the addition of a TV loss further enhances mask manufacturability.

2. Preliminaries

Semiconductor lithography. The primary goal of semiconductor lithography is to accurately model the interaction between mask patterns and wafers through optical projection systems. The mask $M(x, y)$ can be defined as a binary function representing opaque and transparent regions, where (x, y) denotes spatial coordinates on the mask plane. The mask pattern is projected onto the wafer through an optical projection framework H [20, 44], producing an aerial image $I(x', y') = H \cdot M(x, y)$ that represents the light intensity distribution. Here, (x', y') denotes spatial coordinates on the wafer.

The final wafer pattern is determined by the interaction between the aerial image I and the photoresist material on the wafer. This interaction is commonly modeled using a variable threshold resist (VTR) model [35], which captures how light intensity translates into physical features after development. In the VTR model, a dose function $D(x', y') = I(x', y') \otimes G$ is computed, where G represents a spatial kernel encoding the effects of photoresist properties and exposure variability. The resulting resist image R is then obtained as a binary map by applying a dose threshold D_{th} , such as:

$$R(x', y') = \begin{cases} 1, & \text{if } D(x', y') \geq D_{th}, \\ 0, & \text{if } D(x', y') < D_{th}. \end{cases} \quad (1)$$

This overall process from a mask M to the resist image R is represented as a lithography simulator g .

Problem formulation. The *mask optimization* task aims to find an optimal mask M^* that produces a resist image that closely matches a desired target wafer pattern R^* ,

$$M^* = \arg \min_M \mathcal{L}(g(M), R^*), \quad (2)$$

where $\mathcal{L}$ is the loss function measuring the difference between the simulated resist image $R = g(M)$ and the target pattern. Conversely, the *lithography simulation* task focuses on predicting the resist image from a given mask. Accurate simulation is essential not only for validating designs but also for enabling efficient mask optimization, as it serves as the forward model in the optimization loop.

Physics-inspired lithography models. Early DL-based approaches to semiconductor lithography [7, 53, 57] primarily relied on architectures such as UNet++ and CGAN, focusing purely on data-driven learning without incorporating domain-specific physics. However, recent advancements have begun to integrate physics-inspired priors to improve model performance and generalizability. For example, DOINN [54] and CFNO [50] leverage FNOs [23] to implicitly encode optical physics in the frequency domain, demonstrating strong results on both lithography simulation and mask optimization. ILILT [51] takes this further by embedding Inverse Lithography Technology (ILT) [67]—a physics-based framework for mask optimization—directly into the learning process. Nitho [8] introduces a coordinate-based complex MLPs inspired by optical kernel regression [10], specifically designed for lithography simulation. Although these methods embed physics priors, they rely on indirect methods such as frequency-domain modeling or complex MLPs. In contrast, our method explicitly incorporates physical principles into both the architectural design and the core operations, leading to improved generalization and robustness to OOD sets. **The necessity of modeling diffraction in mask optimization** is further examined using Sub-Resolution Assist Features (SRAFs) in Appendix F.

3. Method

3.1. Convolution Operation from Light Diffraction

In wave optics, the propagation of an optical wave U from the aperture plane (x, y) to a target plane (x', y') at distance z can be described by the Rayleigh-Sommerfeld (RS) diffraction integral [41], where λ denotes the wavelength and $k = 2\pi/\lambda$ is the wavenumber.

The RS integral admits several forms, including the Helmholtz–Kirchhoff, the Green’s function, as well as the Fresnel approximation. For clarity of derivation, we illustrate the Fresnel form here, while noting that our framework is fully compatible with other RS-based formulations.

$$U(x', y') = \frac{e^{jkz}}{j\lambda z} \iint_{-\infty}^{\infty} U(x, y) e^{\frac{jk}{2z}[(x'-x)^2 + (y'-y)^2]} dx dy. \quad (3)$$

This integral reveals that the diffraction can be modeled as a convolution with an optical propagation kernel. Recall that the convolution of a function f with a kernel h is mathematically defined as:

$$(f * h)(x', y') = \iint_{-\infty}^{\infty} f(x, y) h(x' - x, y' - y) dx dy. \quad (4)$$

The diffraction integral in Eq. 3 can be rewritten by isolating the quadratic dependence on $(x' - x)$ and $(y' - y)$. This transforms the integral using convolution as follows:

$$\begin{aligned} U(x', y') &= \frac{e^{jkz}}{j\lambda z} \iint_{-\infty}^{\infty} U(x, y) h(x' - x, y' - y) dx dy \\ &= \frac{e^{jkz}}{j\lambda z} [U(x, y) * h(x', y')], \end{aligned} \quad (5)$$

where $h(x', y')$ defines the Optical Phase (OP) kernel that characterizes diffraction through phase modulation. Different RS formulations yield different expressions of the OP kernel, as summarized below (Appendix A for derivations):

$$h(x, y) = \exp\left(\frac{jk}{2z} [x^2 + y^2]\right) \quad (\text{Fresnel approximation})$$

$$h(x, y) = \frac{\exp\left(jk\sqrt{x^2 + y^2 + z^2}\right)}{x^2 + y^2 + z^2} \left[1 + \frac{j}{k\sqrt{x^2 + y^2 + z^2}}\right] \quad (\text{Helmholtz–Kirchhoff})$$

$$h(x, y) = \frac{\exp\left(jk\sqrt{x^2 + y^2 + z^2}\right)}{x^2 + y^2 + z^2} \quad (\text{Green’s function})$$

3.2. OP Kernel-Based Complex Convolution

Building upon the foundation established in Section 3.1, where we demonstrated that the diffraction integral can be expressed as a convolution operation with the OP kernel, we introduce a complex convolution layer that incorporates this physically grounded kernel as a physics-inspired inductive bias. Unlike traditional convolution filters that focus only on spatial features and rely entirely on data-driven learning, our formulation integrates optical priors, which enables the layer to more accurately model wave propagation and thereby enhance lithography performance.

OP kernel construction. For a kernel size of (N, N) , we define the coordinates (x, y) relative to the center of the kernel and compute the complex exponential to construct the OP kernel $Q(x, y) = \exp\left(\frac{jk}{2z} [x^2 + y^2]\right)$. Each element of $Q(x, y)$ corresponds to the phase term of diffraction. In this work, we primarily utilize the OP kernel in Fresnel form due to its effectiveness. However, our framework is formulation-flexible and can also incorporate other alternative RS formulations, as well as the Hopkins transmission cross-coefficient (TCC) kernel. The pseudo-code for our OP kernel is provided in Appendix C.

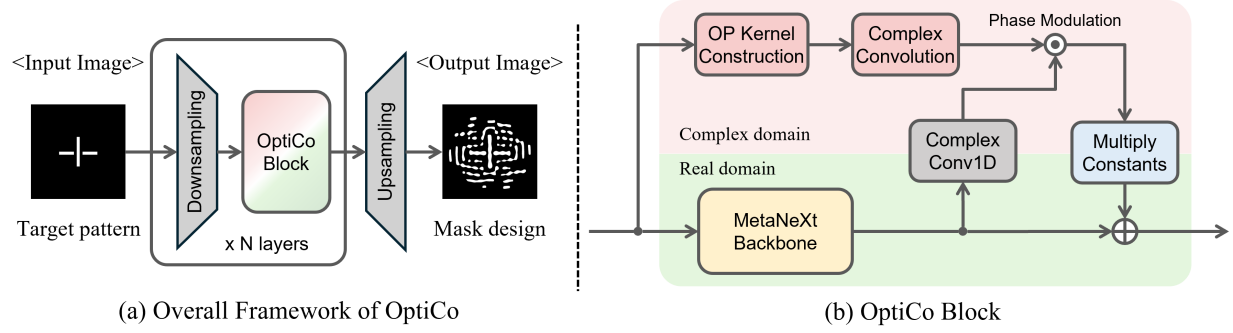


Figure 2. Overview of the OptiCo framework and the design of its block integrating OP kernel.

OP complex convolution. To integrate the diffraction-inspired OP kernel Q into our architecture, we can construct an effective kernel W_{eff} in two possible forms: either element-wise multiplication with a learnable complex weight W , or a stricter variant scaled only by a learnable scalar weight λ_α :

$$W_{\text{eff}} = Q \odot W \quad \text{or} \quad W_{\text{eff}} = \lambda_\alpha \cdot Q, \quad (6)$$

where $\odot$ denotes element-wise multiplication. The resulting complex effective kernel W_{eff} encodes optical diffraction, so that each convolution step inherently reflects the physical propagation of light.

Since optical fields are inherently complex, we decompose input U and effective kernel W_{eff} into real (r) and imaginary (i). The OP complex convolution is defined as:

$$U = U_r + jU_i; \quad W_{\text{eff}} = W_r + jW_i; \quad (7)$$

$$\begin{aligned} \text{OP}_{\text{conv}}(U) = & \left[(U_r * W_r - U_i * W_i) \right. \\ & \left. + j(U_r * W_i + U_i * W_r) \right]. \end{aligned} \quad (8)$$

3.3. Architectural Design of the OptiCo Block

The OptiCo block integrates a MetaNeXt-based backbone [61] with the OP complex convolution in a unified layer. Our backbone block takes input U as:

$$\begin{aligned} Y_{\text{backbone}}(U) = & \left(\text{DWConv}(\text{Norm}(U_r)W_1) \right. \\ & \left. \odot \sigma(\text{Norm}(U_r)W_2) \right) W_3 + U_r, \end{aligned} \quad (9)$$

where DWConv is a depthwise convolution, σ an activation function, and W_1, W_2, W_3 are the MLP parameters. Since the OP convolution is inherently complex, we first embed Y_{backbone} into the complex domain using a per-pixel ComplexConv1D (CCConv1D), yielding a complex feature. At each coordinate (p, q) , the channel vector $x_{pq} \in \mathbb{R}^C$ is embedded into the complex domain through projections:

$$\text{CCConv1D}(x_{pq}) = (x_{pq,r}V_r - x_{pq,i}V_i) + j(x_{pq,r}V_i + x_{pq,i}V_r), \quad (10)$$

where $V_r, V_i \in \mathbb{R}^{C \times C}$ are learnable projection matrices. This operation serves as a per-pixel channel embedding, producing complex-valued features that can interact with the OP kernel.

This embedding is combined with OP_{conv} via Hadamard product that performs phase modulation. The phase modulated output Y_{phase} is multiplied by the constants and finally added back to the backbone feature, injecting physically grounded diffraction effects into the mask representation.

$$Y_{\text{phase}}(U) = \frac{e^{jkz}}{j\lambda z} \left[\text{CCConv1D}(Y_{\text{backbone}}(U)) \odot \text{OP}_{\text{conv}}(U) \right], \quad (11)$$

$$Y_{\text{OptiCo}} = Y_{\text{backbone}}(U) + Y_{\text{phase}}(U). \quad (12)$$

3.4. Total Variation Loss for Unwanted Mask Artifact Suppression

To account for practical constraints in mask production, we aim to not only minimize the discrepancy between the target wafer pattern R^* and the simulated resist image $g(M)$ from the optimized mask M , but also enforce a regularization loss term to suppress *grainy* artifacts that hinder mask manufacturability. The primary loss is formally defined as:

$$\mathcal{L}_{\text{mse}}(M) = \|g(M) - R^*\|^2, \quad (13)$$

where $g(\cdot)$ denotes the lithography simulator, which generates the corresponding resist image from a given optimized mask. To promote spatial smoothness in the mask and suppress high-frequency noise, we introduce a standard 2D total variation (TV) loss [63]:

$$\mathcal{L}_{\text{tv}}(M) = \sum_{p,q} \left| M_{p+1,q} - M_{p,q} \right| + \left| M_{p,q+1} - M_{p,q} \right|. \quad (14)$$

This loss term encourages spatial smoothness by penalizing abrupt changes between adjacent pixels, thereby reducing unwanted *grainy* artifacts. Combining these two terms yields our final objective:

$$\mathcal{L}_{\text{final}}(M) = \mathcal{L}_{\text{mse}}(M) + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}}(M), \quad (15)$$

where λ_{tv} is the weighting parameter that controls the strength of the regularization.

Table 1. Comparative performance on mask optimization across competing methods. Lower is better.

Method	MetalSet			ViaSet			StdMetal (MetalSet OOD)			StdContact (ViaSet OOD)			Average		
	MSE	PVB	EPE	MSE	PVB	EPE	MSE	PVB	EPE	MSE	PVB	EPE	MSE	PVB	EPE
DAMO	32579	41173	5.4	5081	9962	0.0	16120	23796	0.2	50445	35673	26.7	26056	27651	8.1
DOINN	36409	41929	7.4	4382	7836	0.0	25913	25749	4.5	72058	17968	55.8	34691	23370	16.9
CFNO	47814	46131	12.5	8949	9890	0.1	26809	26814	4.2	70740	17950	55.1	38578	25196	18.0
ILILT	30353	45353	3.2	4666	10065	0.0	14596	24969	0.1	38957	43869	7.1	22143	31064	2.6
OptiCo	24033	45379	1.6	4339	7802	0.0	11293	25183	0.0	18474	39181	0.1	14535	29373	0.4

4. Experiments

4.1. Experimental Design and Configurations

Dataset. To evaluate the performance of our OptiCo framework, we utilize LithoBench [65], which is one of the most recent and widely recognized computational lithography benchmarks. Specifically, LithoBench provides over 100,000 layout tiles, consisting of four subtasks: two synthetic design sets (MetalSet, ViaSet) and two real-world sets treated as OOD (StdMetal, StdContact). Each subtask includes a variety of data types such as target images, reference masks, aerial images, and resist images. The resolution of each image is 2048×2048 pixels, with each pixel representing a $1nm^2$ of the semiconductor layout.

Setup. In this work, we conducted experiments primarily focused on mask optimization, where the model receives target designs as input and generates optimized masks as output. This task is essential for ensuring that the resulting lithographic masks enable the precise transfer of circuit layouts onto silicon wafers. In addition to mask optimization, we carried out an additional experiment on lithography simulation. In this task, the model takes optimized masks as input and produces the corresponding aerial and resist images as output. The primary objective of lithography simulation is to accurately predict the final wafer patterns, ensuring that the generated resist images closely align with the intended circuit layouts. We closely follow the original LithoBench hyperparameter configuration and training procedure, and the full hyperparameter settings and experimental details are provided in Appendix E.2.

Evaluation metrics. For the mask optimization task, we used three evaluation metrics: mean squared error (MSE), edge placement error (EPE) violations, and process variation band (PVB) area. Specifically, MSE measures the overall discrepancy between the simulated resist image and the target wafer pattern. EPE and PVB are industry-standard metrics commonly used in chip manufacturing to evaluate mask quality. EPE quantifies the deviation between the intended edge positions on the resist image and their actual positions. PVB measures the range of possible edge positions caused by manufacturing process variations. For the lithography simulation task, the performance of each

method was assessed using two evaluation metrics: MSE and intersection over union (IoU). Detailed explanations of these evaluation metrics are provided in Appendix E.1.

Competing methods. We compared our OptiCo framework against several prominent semiconductor lithography models from the LithoBench, as well as recent specialized models. In particular, DAMO [7] is a fully data-driven model that uses a UNet++ [66] backbone. In addition, DOINN [54] and CFNO [50] leverage FNO-based architectures that implicitly model diffraction effects, making them the most relevant comparative baselines. They transform the input image with a Fourier transform, embed it via MLP layers, and then concatenate those embedded features with local CNN outputs. Furthermore, ILILT [51] is a recent state-of-the-art mask optimization model that iteratively inputs simulated resist images along with target patterns, and Nitho [8] is a physics-inspired lithography simulation model based on optical kernel regression.

4.2. Experimental Results

Mask optimization experiments. Table 1 provides a comprehensive evaluation of all methods across different subtasks in the mask optimization task. Our OptiCo framework consistently achieves superior performance across most evaluation metrics and subtasks, highlighting the effectiveness of integrating optical diffraction principles into the neural network architecture. Notably, OptiCo demonstrates exceptional performance on the StdMetal and StdContact subtasks, which are inherently more challenging and demand high generalization capabilities for OOD tasks.

Strengths in OOD. In the ViaSet, nearly all methods achieve an EPE of zero, indicating strong adaptation to the training dataset. However, on the OOD test set (StdContact), most competing methods exhibit a sharp increase in EPE. In contrast, our OptiCo consistently maintains a robust performance on the OOD test set. This robustness further highlights the effectiveness of incorporating optical diffraction principles, which offer physics-inspired guidance and enhance the model’s generalization capability. Among competing methods, ILILT demonstrates strong performance. However, our OptiCo surpasses ILILT by explicitly incorporating physical principles via the OP kernel.

Table 2. Lithography simulation performance across competing methods. Lower MSE is better and higher IoU is better.

Method	MetalSet			ViaSet			StdMetal (MetalSet OOD)			StdContact (ViaSet OOD)			Average		
	MSE _{aerial}	MSE _{resist}	IoU	MSE _{aerial}	MSE _{resist}	IoU	MSE _{aerial}	MSE _{resist}	IoU	MSE _{aerial}	MSE _{resist}	IoU	MSE _{aerial}	MSE _{resist}	IoU
DAMO	8.36e-6	7.48e-4	0.97	2.98e-6	1.47e-4	0.94	2.50e-5	1.50e-3	0.96	4.56e-5	1.64e-3	0.87	2.05e-5	1.01e-3	0.93
DOINN	9.45e-6	6.77e-4	0.97	2.02e-6	1.04e-4	0.96	1.36e-5	1.29e-3	0.96	2.72e-5	1.37e-3	0.89	1.31e-5	8.61e-4	0.95
CFNO	1.87e-5	1.47e-3	0.94	3.77e-6	2.13e-4	0.92	2.61e-5	2.29e-3	0.93	2.14e-5	2.20e-3	0.83	1.75e-5	1.54e-3	0.90
Nitho	9.27e-6	7.95e-4	0.97	2.24e-6	1.46e-4	0.95	2.80e-5	1.94e-3	0.95	2.22e-5	1.82e-3	0.85	1.54e-5	1.18e-3	0.93
OptiCo	2.44e-6	4.18e-4	0.98	1.04e-6	8.88e-5	0.96	7.19e-6	8.59e-4	0.98	1.01e-5	1.24e-3	0.90	5.01e-6	6.38e-4	0.96

Table 3. Ablation of the OP kernel across alternative diffraction formulations to assess formulation-flexible behavior.

Diffraction formulas	StdMetal (EPE)	StdContact (EPE)
w/o kernel	2.819	22.612
Helmholtz	0.207	0.303
Hopkins	0.125	0.461
Green’s	0.059	0.188
Fresnel	0.044	0.079

Table 4. Ablation of the strict OP kernel variant $W_{\text{eff}} = \lambda_{\alpha} Q$, which more faithfully reflects high-fidelity diffraction models.

Diffraction formulas	StdMetal (EPE)	StdContact (EPE)
w/o kernel	2.819	22.612
Helmholtz	0.786	5.497
Hopkins	0.044	0.212
Green’s	0.059	0.309
Fresnel	0.173	1.061

Table 5. An ablation study for each component in OptiCo block: OP complex convolution (OP), Multiply Constants (MC), ComplexConv1D (CC).

Ablation study			StdMetal (EPE)	StdContact (EPE)
OP	MC	CC		
-	-	-	2.819	22.612
-	-	✓	0.657	7.491
✓	-	-	1.561	6.321
✓	✓	-	0.188	1.273
✓	✓	✓	0.044	0.079

Additional experiments. Table 2 presents the performance evaluation for the lithography simulation. The evaluation metric MSE_{resist} and MSE_{aerial} represent the MSE losses for the resist and aerial images. Consistent with our main findings, OptiCo outperforms all competing methods across subtasks, underscoring the robustness of OptiCo across diverse tasks. Furthermore, to evaluate the benefit of embedding optical physics, we performed a train **dataset-ratio ablation** in Appendix B.3, comparing physics-aware and purely data-driven methods. To investigate the compatibility of OptiCo, we conducted experiments with **different backbone networks**, such as DOINN and CFNO in Appendix B.2 and a **low resolution setting** in Appendix B.4, showing that OptiCo consistently adapts well across architectures and different resolution scales.

4.3. Ablation Study

Alternative diffraction kernel. The OP kernel is designed to embed diffraction principles into the network in a formulation-flexible manner. To examine this, we replaced the default Fresnel kernel with several alternative diffraction formulations. In Table 3, the OP kernel remains effective across these variants relative to the no-kernel baseline. Moreover, the simpler Green’s and Fresnel kernels achieve the best performance, indicating that lightweight formulations integrate more naturally with neural architectures.

Strict OP kernel variant. In Table 4, we examine a stricter effective kernel variant in which the OP kernel is scaled only by a scalar weight. This setting amplifies the importance of the diffraction model’s physical fidelity: the Hopkins TCC kernel achieves the highest performance, consistent with its role as the widely used standard

in computational lithography and indicating that our framework aligns well with high-fidelity physics models. While the paraxial Fresnel approximation experiences a noticeable drop, the non-paraxial Green’s formulation, which has better numerical aperture (NA) suitability, remains robust.

Advantage of modeling in a complex domain. In Table 5, we assess the contribution of each component in the OptiCo block. Since the phase is a crucial component of light, modeling in the complex domain is necessary. To validate this, we ablated OP complex convolution (OP), applying only ComplexConv1D (CC) in OptiCo. Even this simple modification leads to a notable performance gain, confirming the effectiveness of complex-domain modeling.

Completeness of phase modulation. In the OptiCo block, multiplying constants (MC) to OP are essential for preserving the complete diffraction formulation. When MC are ablated from OP, we observe a substantial performance drop, highlighting the importance of incorporating all formulation components. Finally, adding CC to OP completes the phase modulation and further enhances performance, underscoring the benefit of a faithful architectural design.

Comparison with FNO. While OptiCo explicitly incorporates diffraction principles, the Fourier Neural Operator (FNO) serves as a strong baseline, as its Fourier modeling can implicitly capture diffraction principles. To examine this distinction, we replaced OptiCo block with an FNO and compared the performance. As shown in Figure 3, substituting FNO yields a clear performance gain, consistent with our earlier finding on the importance of complex-domain modeling. Nevertheless, OptiCo achieves superior performance by explicitly embedding diffraction principles. Additional evaluation metrics are provided in Appendix B.1.

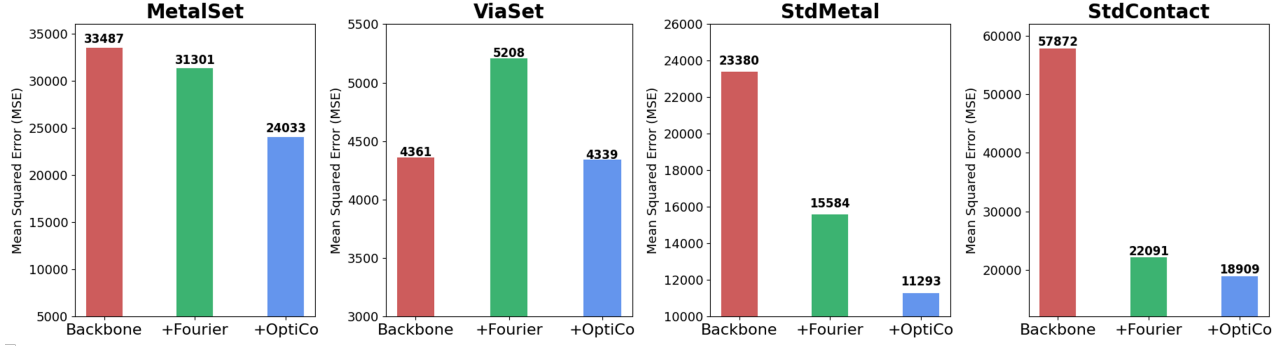


Figure 3. Comparative performance of the backbone network with the addition of Fourier module (FNO) and our OptiCo.

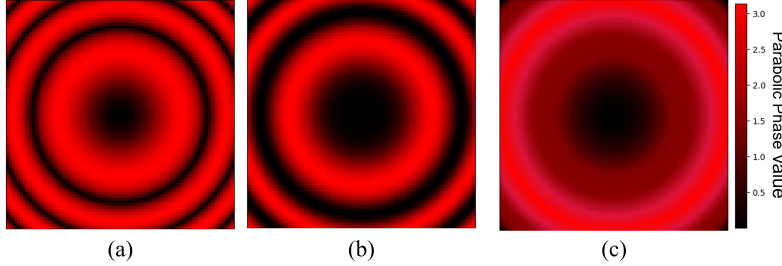


Figure 4. Visualization of the OP kernel across dynamic wavefront scales.

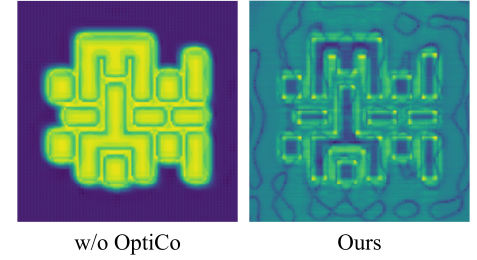


Figure 5. Intermediate feature visualization.

Kernel size ablation. A kernel that is too small results in a limited receptive field, making it insufficient to capture a diffraction, while a large kernel introduces complexity. In Appendix B.5, performance initially improves as the kernel size increases, but degrades when it becomes too large. We attribute this to the quadratic nature of a kernel ($x^2 + y^2$), as steep phase variations hinder training. Nevertheless, OptiCo consistently outperforms strong baselines (ILILT).

4.4. Diffraction-Inspired Representations

Visualization of the OP kernel. To provide an intuitive understanding of how the OP kernel encodes diffraction principles, we visualize its spatial structure across different wavefront scales. As illustrated in Figure 4, the images represent distinct configurations of the OP kernel, $\exp\left(\frac{jk}{2z}[x^2 + y^2]\right)$, each illustrating its dynamic adaptation to varying wavefront parameters. The accompanying color map on the right quantitatively represents parabolic phase values, where darker shades correspond to lower phases and brighter reds indicate higher accumulations. A key observation across all images is the presence of concentric rings with alternating intensity—an intrinsic characteristic of the parabolic phase term in diffraction. These rings indicate phase variations that reflect the OP kernel’s capacity to accurately model light propagation. Comparing images (a) through (c), we observe dynamic changes in the spacing, sharpness, and intensity of rings. Specifically, in (a), the rings are tightly packed, indicating high-frequency phase variations, while in (b) and (c), the rings are more widely spaced, reflecting a lower-frequency phase structure.

Visualization of intermediate feature. To further examine how the OP kernel influences internal representations, we provide intermediate feature visualizations comparing models with and without OptiCo in Figure 5. These results show that OptiCo focuses on diffraction-sensitive regions—particularly pattern corners and surrounding sub-resolution assist feature (SRAF) areas—highlighting OptiCo’s ability to capture optical principles. In summary, these visual representations underscore the OP kernel’s capacity to accurately model the complex optical lithography. Additional examples are provided in Appendix D.2.

Visualization of diffraction-inspired mask patterns. We compared the mask patterns optimized by OptiCo with those from other competing methods in Figure 6. Notably, OptiCo produces mask patterns featuring more distinct outer ring structures than its competitors. These diffraction-inspired ring structures arise from integrating light diffraction principles through the OP kernel, as shown in Figure 4. In particular, this phenomenon is especially pronounced when comparing ViaSet with its OOD test set, StdContact. OptiCo produces highly structured mask patterns in both datasets, clearly guided by the OP kernel’s explicit modeling of light diffraction. In contrast, DOINN, which is an FNO-based framework, forms complex diffraction patterns on the ViaSet but fails to replicate them on the StdContact (OOD test set). This discrepancy underscores the advantage of explicitly incorporating optical diffraction principles into semiconductor lithography through the use of our OP kernel. Additional visualizations for all methods and other subtasks are provided in Appendix D.1.

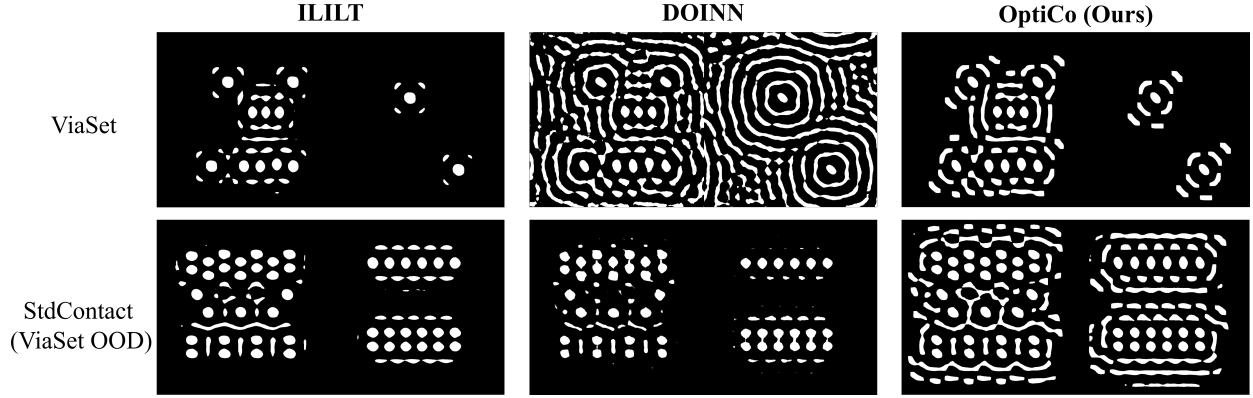


Figure 6. Visualization of optimized mask patterns. OptiCo produces clearer diffraction-inspired rings, especially on StdContact, highlighting the benefits of explicit diffraction modeling. In contrast, the Fourier-based DOINN shows complex rings on ViaSet but fails to preserve them on StdContact.

4.5. Manufacturability Improvement with TV Loss

Artifact suppression. In mask optimization, *grainy artifacts* refer to small, high-frequency variations or speckles present in the mask pattern. While these artifacts may appear insignificant, they present substantial challenges in practical manufacturing, as they can lead to unpredictable deviations in the final pattern on the wafer. As shown in Figure 7, we compare the grainy artifacts in optimized mask patterns with and without the total variation (TV) loss. The model without TV loss produces noticeable grainy artifacts (highlighted in red), which pose practical challenges in mask manufacturing. In contrast, integrating TV loss into our model largely suppresses these grainy artifacts, resulting in smoother boundaries and more coherent shapes. This highlights the advantage of TV loss in enhancing the manufacturability of the masks.

Simplified mask and post-processing robustness. We also assessed manufacturability by comparing shot counts, which reflect the complexity of the layout geometry. As in Table 6, incorporating TV loss leads to a slight degradation in EPE but a substantial improvement in shot count, reducing it to less than half of that without TV loss. For further analysis, we applied morphological post-processing to the mask images [52], a commonly used technique that suppresses artifacts and smooth boundaries, though it inevitably degrades performance. Intuitively, if the mask is clean, the performance drop after post-processing should remain small. Indeed, masks trained with TV loss exhibit only minor degradation in EPE and negligible changes in shot count, confirming that TV loss promotes cleaner masks.

5. Conclusion

In this paper, we propose OptiCo, a novel physics-inspired convolutional neural network that integrates diffraction principles directly into its architecture to enhance computational lithography. Unlike conventional approaches that

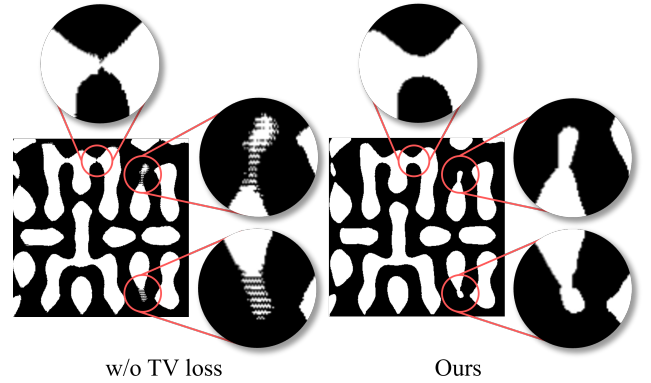


Figure 7. Comparison of mask patterns with and without the total variation (TV) loss, highlighting reduced grainy artifacts and smoother boundaries with coherent shapes in red circle.

Table 6. TV loss produces cleaner masks, leading to minimal EPE degradation and stable shot count after post-processing (PP).

Loss function	EPE ($\downarrow$)	Shot ($\downarrow$)
$\mathcal{L}_{\text{mse}}$	1.4	1523
$\mathcal{L}_{\text{mse}} + \text{PP}$	2.0 ($\Delta+0.6$)	1451 ($\Delta-72$)
$\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{tv}}$	1.6	715
$\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{tv}} + \text{PP}$	1.8 ($\Delta+0.2$)	712 ($\Delta-3$)

rely solely on data-driven methods or Fourier-based frameworks, OptiCo introduces a physics-inspired OP kernel to explicitly model the near-field diffraction behavior of light. Empirical evaluations on well-established lithography benchmarks demonstrate that our OptiCo consistently outperforms existing state-of-the-art methods in both lithography simulation and mask optimization tasks. Moreover, our in-depth analysis underscores the crucial role of the OP kernel in explicitly capturing the parabolic phase variations inherent to diffraction principles.

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